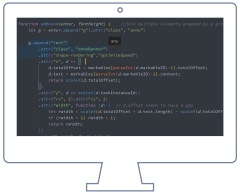
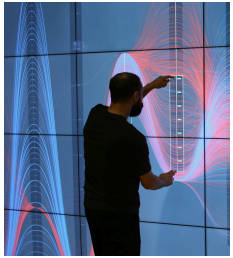
Information Visualization Course

Patrick Riehmann

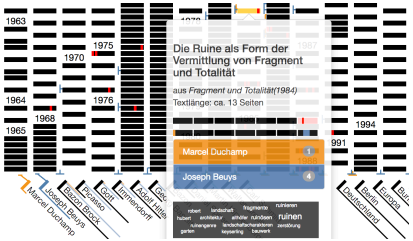
Who am I



Software Development



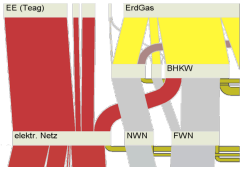
HCI



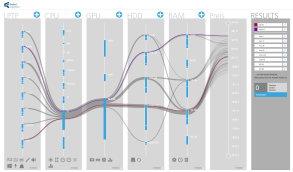
Information Visualization

Web Technology

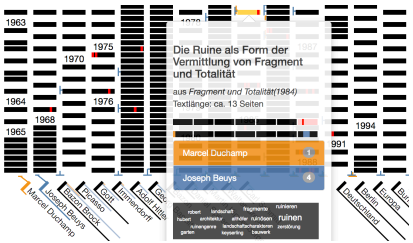
Contributions and Research Topics (see riehmann.net)



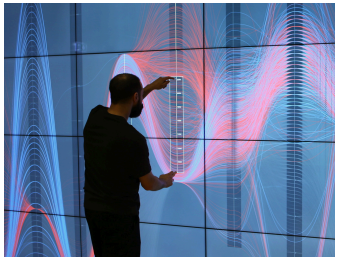
Diagrams and Graph Drawing



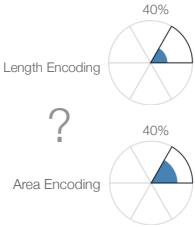
Configuration/Decision Support



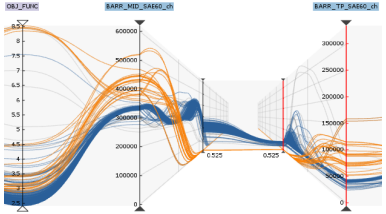
Text-based Visualization



Direct Touch Analytics



Perception and Encoding



Multivariate Data and Time Series

Course Overview

- Video lectures
- Live Questions & Comprehension sessions (Q+A, weekly) all questions concerning:
  - Details of the video lecture topics
  - Details and specifics of the assignments
  - Details of the lab work
  - All organizational matters
  - Exam specifics
- Assignments
  - 5 theoretical assignments
  - 2 practical assignment
  - Missed or failed assignments are to be re-submitted during the next re-examination week (see academic calendar)
- Lab class (not mandatory, but recommended):
  - Introduction into the Observable Plot Framework (preparing for the practical assignments 6-7)
  - Time for doing the theoretical and practical assignments with support from lab assistants
- Exam

What do I need to pass?

➡ 50% of everything (each theoretical assignment, each practical assignment, exam)

Course Week	1	2	3	4	5	6	7
Calendar Week	44	45	46	47	48	49	50

Lecture Topics (mainly video lectures)

	(00) Intro	(04) Multi-Attributes	(06) Sets	(08) Graphs	(10) Glyphs		
	(01-03) What-Why-How	(05) Parallel Coordinates	(07) Trees	(09) Time Series	(11) Text-based		

Q+A weekly live session with all questions concerning contents of the video lectures, details of the assignments, organizational matters and exam specifics

	Q+A 1	Q+A 2	Q+A 3	Q+A 4	Q+A 5	Q+A 6	Q+A 7

Assignments (Submissions by Sunday 23:59 in the respective week)

	Theoretical Assignments					Practical Assignments	
	1: Your Data (Before videos 01-03)	2: WWH	3: Multi-Attributes	4: MA/S/T	5:Vis Analytics	6: Plot1	6:Plot2 (Your Data)

Labs Week 2 - 5: (Labs are not mandatory, but strongly recommended)
- 2 introduction into the Observable Plot Framework (preparing for the practical assignments 6-7)
- 2 hours time for the doing the theoretical assignments

Labs Week 6 - 7:
- 4 hours time for the doing the practical assignments

	No lab	Plot Intro I	Plot Intro II	Plot Intro III	Plot Intro IV	Practical Assigment	Practical Assigment

Organisation: First and Second Week

First Week:

(1) CYOD - Create your own data:
The first assignment is an online user study to create your own data for the course.
Do this before watching the video lectures! It only takes 10 minutes.

→ Yes, **before** the videos!!!

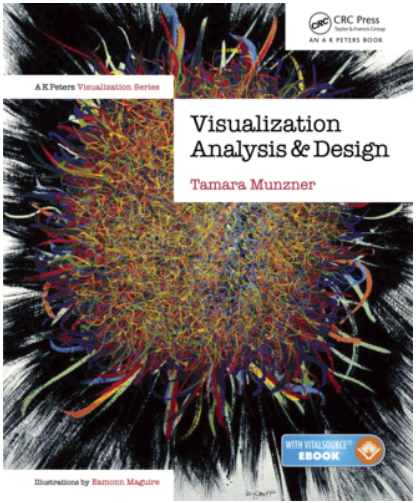
(2) Watch video lectures 01-03 this week, since you need them for second weeks assignment!

→ Yes, **three** video lectures, since you have **no labs** in the **first week**!

Second Week:

- Do the second assignment sheet
- Watch video lecture 04 and 05
- Recommended: Go to the lab for the first part of the introduction into Observable Plot (90 minutes)!
- After that: Feel free to ask the lab assistants for hints concerning the assignments!
- If you have any questions concerning content, assignments or organizational matters feel free to attend the Q+A session.

Further Readings

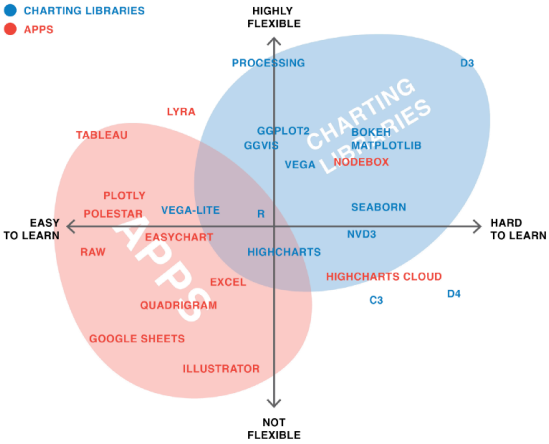


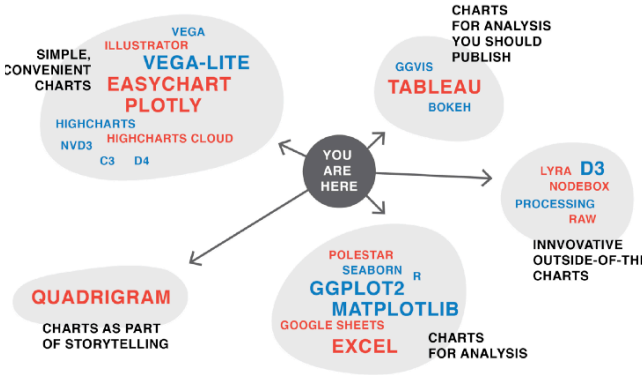
If material is not explicitly referenced in the slides its from his book!

Two big themes (according to Tamara Munzner)

- Foundations (Theory)
  - Helps in understanding tradeoffs of design and how to make informed decisions
    - Principles: Should I design in this or that way
    - Techniques: What kinds of design are possible
- Tooling mechanics / programming
  - How to build a visualization
    - Application-based (Excel, Tableau, Illustrator)
    - Coding (D3, Javascript, Python, etc.)

Tools, Tools, Tools, …





https://source.opennews.org/articles/what-i-learned-recreating-one-chart-using-24-tools/

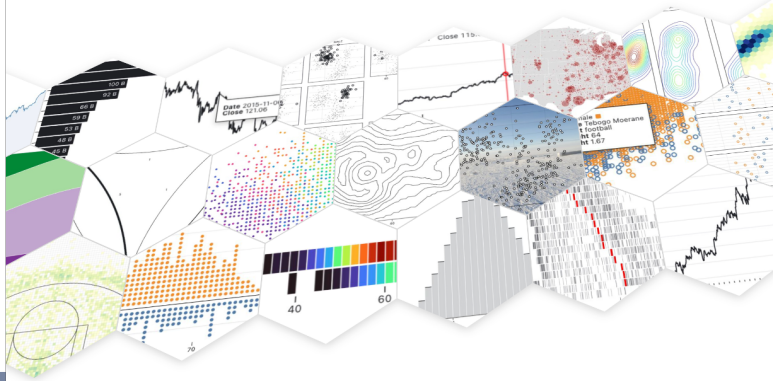
Practical Assignments

Observable Plot

The JavaScript library for exploratory data visualization

Create expressive charts with concise code

Get started | What is Plot? | Examples



The Visualization Zoo

- Time-Series Data
- Statistical Distributions
  Multi-variate Data
- Map-embedded Data
- Hierarchies
- Networks
- (Set-based Data)

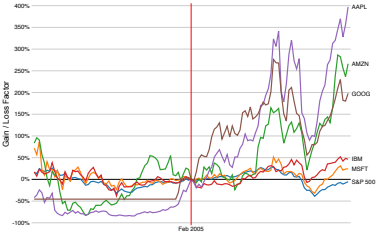
A survey of powerful visualization techniques, from the obvious to the obscure.

BY JEFFREY HEER, MICHAEL BOSTOCK, AND VADIM OGIEVETSKY

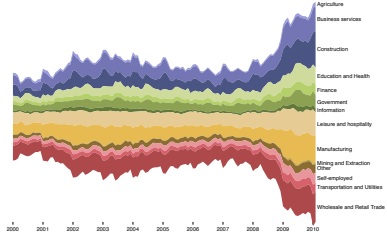
A Tour Through the Visualization Zoo

THANKS TO ADVANCES in sensing, networking, and data management, our society is producing digital information at an astonishing rate. According to one estimate, in 2010 alone we will generate 1,200 exabytes—60 million times the content of the Library of Congress. Within this deluge of data lies a wealth

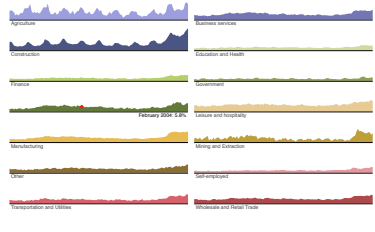
Time Series



Index Charts



Stacked Graphs



Small Multiples

Braided Graphs

Horizon Graphs

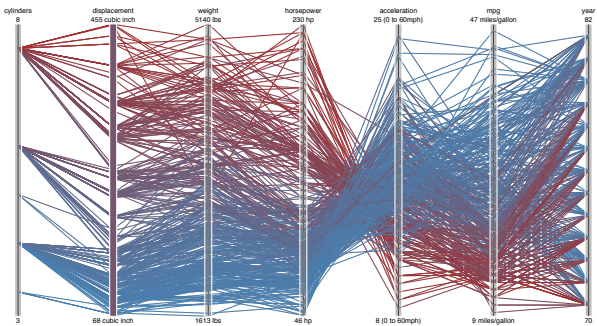
Tabular Data / Multi-Variate Data

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box		7/17/07
32	7/16/07	2-High	Medium Box		7/18/07
32	7/16/07	2-High	Medium Box		7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
69	6/6/05	4-Not Specified	Small Pack	0.44	6/6/05
69	6/6/05	4-Not Specified	Wrap Bag	0.6	6/6/05
70	12/18/06	5-Low	Small Box	0.59	12/23/06

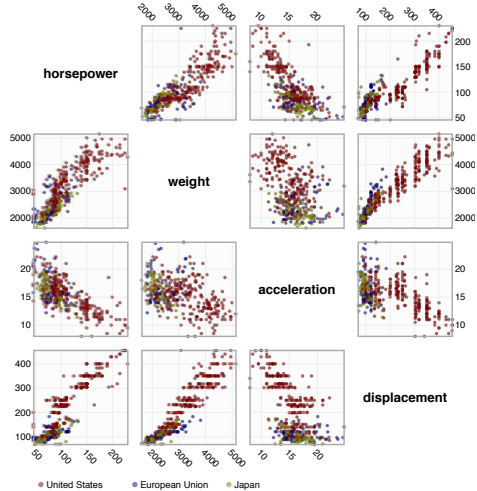
attribute

item

cell



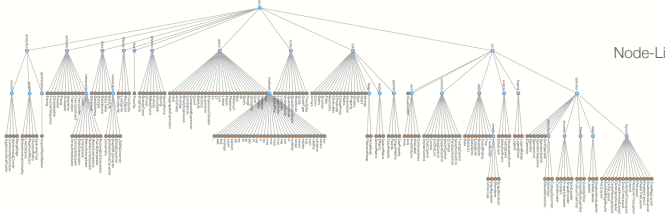
Parallel Coordinates



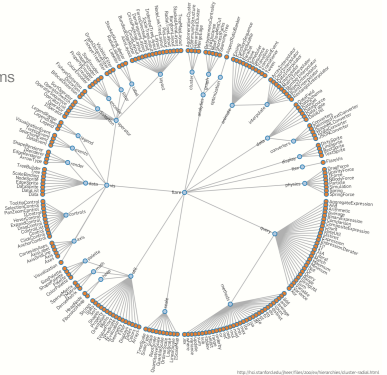
Scatterplots & Scatterplot Matrices

Hierarchies / Trees

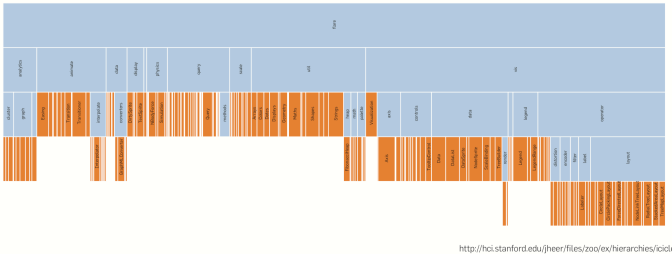
Node-Link Diagrams



http://hci.stanford.edu/jheer/files/zoo/ex/hierarchies/tree.html



http://hci.stanford.edu/jheer/files/zoo/ex/hierarchies/cluster-radial.html



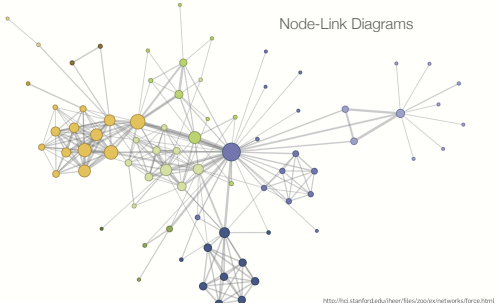
http://hci.stanford.edu/jheer/files/zoo/ex/hierarchies/icicle.html

http://hci.stanford.edu/jheer/files/zoo/ex/hierarchies/treemap.html

Space filling techniques

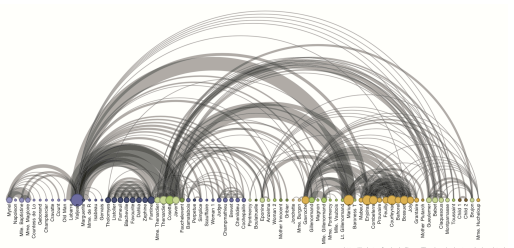
Node-Link Diagrams / Graphs

Node-Link Diagrams

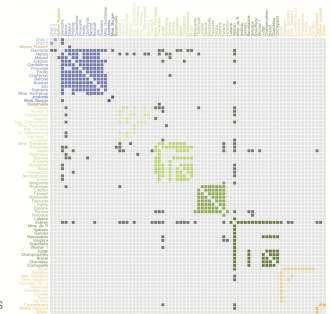


http://hci.stanford.edu/jheer/files/zoo/ex/networks/force.html

Arc Diagrams



http://hci.stanford.edu/jheer/files/zoo/ex/networks/arc.html

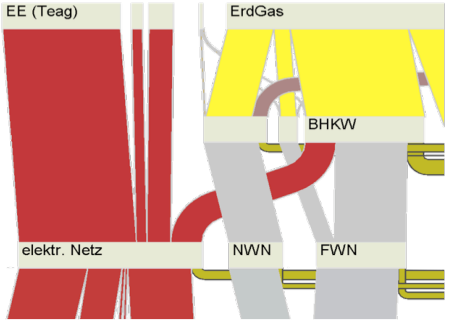


Adjacency Matrices

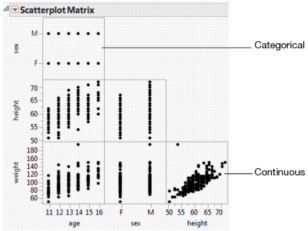
http://bost.ocks.org/mike/miserables/

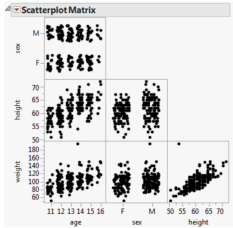
Source: http://www.semanticarts.com/2006/05/07/publishing_the_enron_email_archive/

Sankey Diagrams



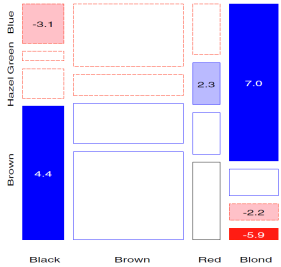
Set-based Data or Categorical Data



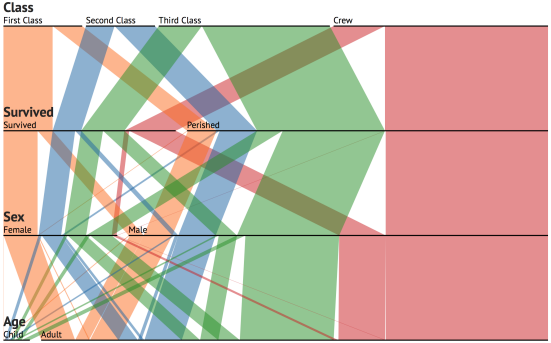


http://www.jmp.com/support/help/Example_of_a_Scatterplot_Matrix.shtml

Jittering Techniques

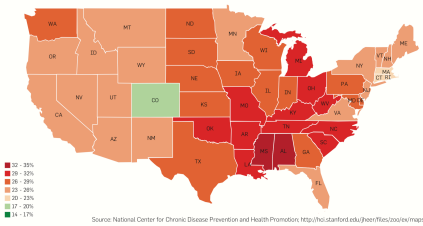


Mosaic Charts



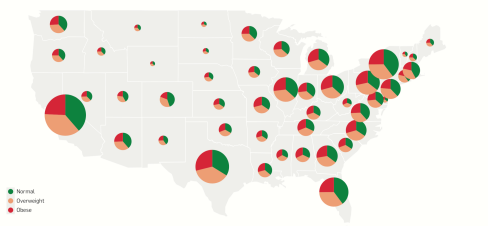
Parallel Sets

Map-based Data



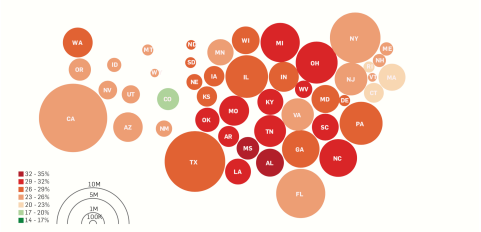
Source: National Center for Chronic Disease Prevention and Health Promotion; http://hci.stanford.edu/jheer/files/zoo/ex/maps/choropleth.html

Choropleth Maps



Source: National Center for Chronic Disease Prevention and Health Promotion; http://hci.stanford.edu/jheer/files/zoo/ex/maps/symbol.html

Graduated Symbols



Source: National Center for Chronic Disease Prevention and Health Promotion; http://hci.stanford.edu/jheer/files/zoo/ex/maps/cartogram.html

Dorling Cartograms

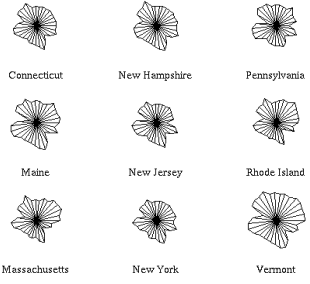
Glyphs & Icons

Direct-Metaphorical Icons

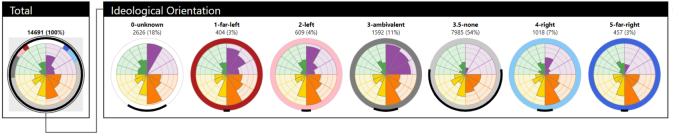
Fat Fonts

Chernoff Faces

Star Glyphs

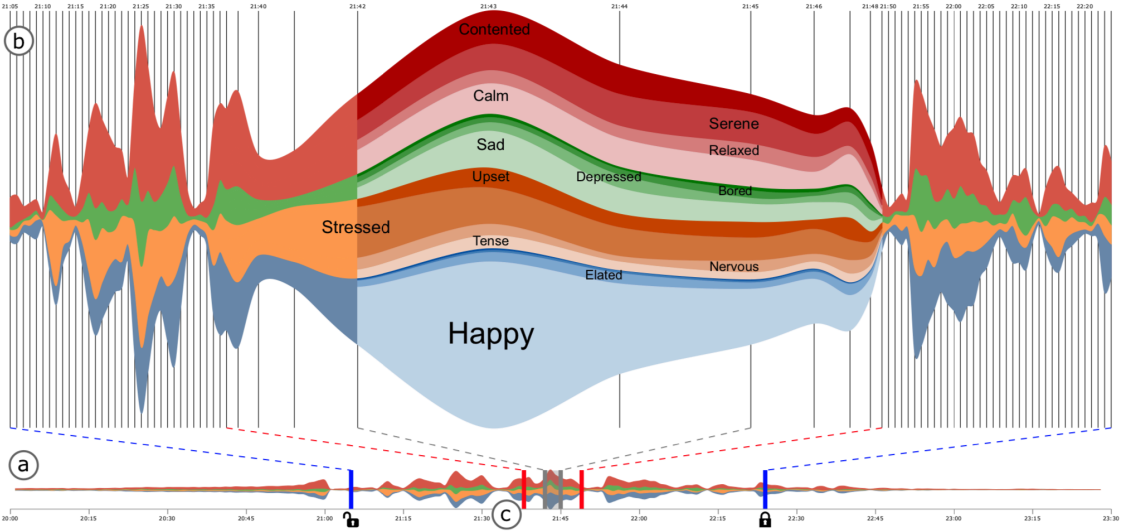


Compound Glyphs



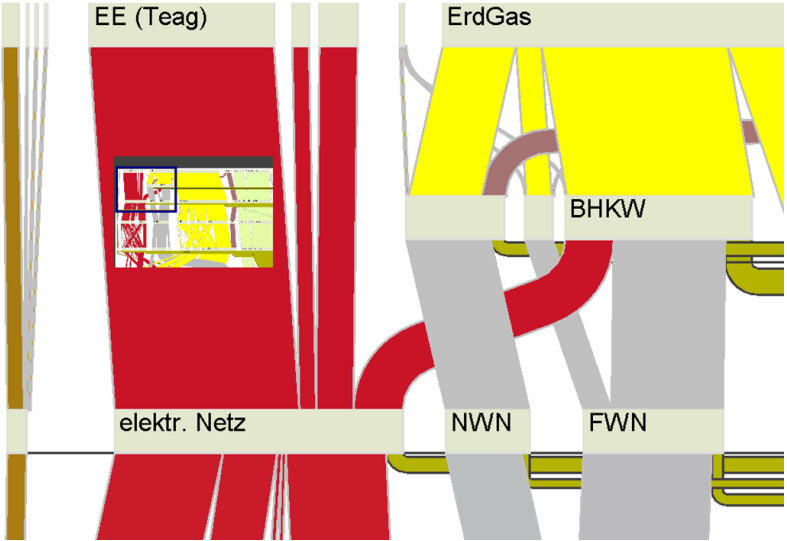
General
InfoVis Concepts/Paradigms and Techniques

Focus and Context

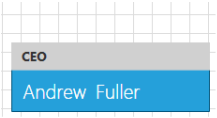


Cuenca, Erick & Sallaberry, Arnaud & Wang, Florence & Poncelet, Pascal.
Visualizing Hierarchical Time Series with a Focus+Context Approach.

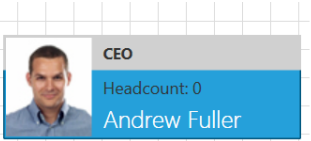
Detail + Overview



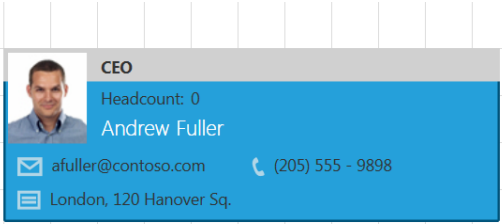
Semantic Zoom



ZoomFactor = 1

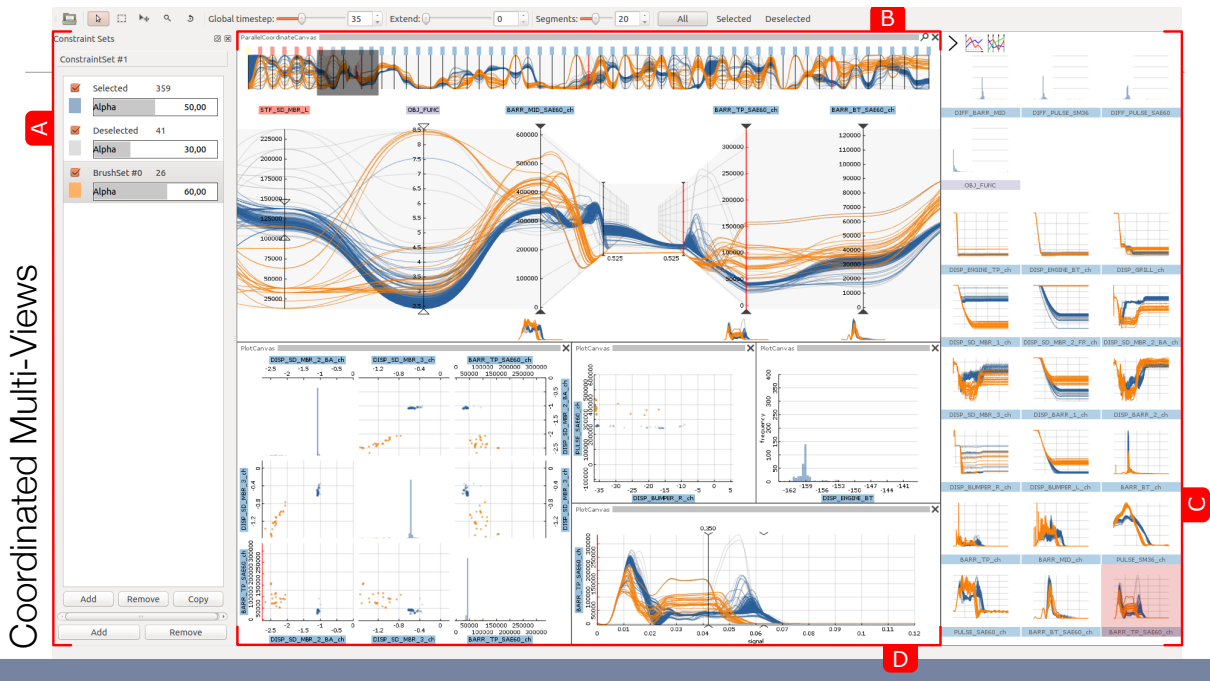


ZoomFactor = 1.5



ZoomFactor = 2.5

https://docs.telerik.com/devtools/wpf/controls/raddiagram/howto/semantic-zoom

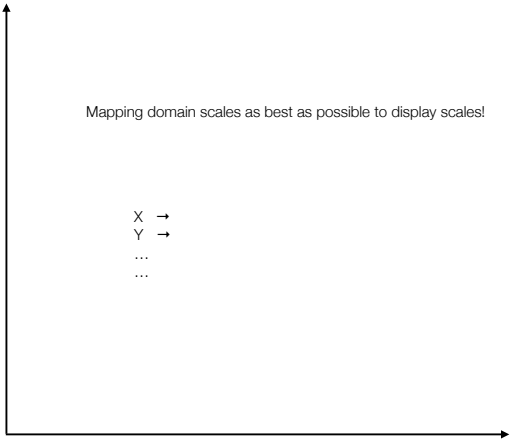

Coordinated Multi-Views
A
B
C
D
Global timestep:
Extend:
Segments:
All
Selected
Deselected
Constraint Sets
ConstraintSet #1
Selected 359
Alpha 50,00
Deselected 41
Alpha 30,00
BrushSet #0 26
Alpha 60,00
Add
Remove
Copy
Add
Remove
ParallelCoordinateCanvas
STF_SD_MBR_L
OBJ_FLUC
BARR_MID_SAE6O_ch
BARR_TP_SAE6O_ch
BARR_BT_SAE6O_ch
PlotCanvas
DISP_SD_MBR_2_BA_ch
DISP_SD_MBR_3_ch
BARR_TP_SAE6O_ch
DISP_SD_MBR_2_FR_ch
DISP_SD_MBR_1_ch
DISP_BUMPER_R_ch
DISP_ENGINE_BT
BARR_TP_SAE6O_ch
signal
frequency
DIFF_BARR_MID
DIFF_PULSE_SM36
DIFF_PULSE_SAE6O
OBJ_FLUC
DISP_ENGINE_TP_ch
DISP_ENGINE_BT_ch
DISP_SPILL_ch
DISP_SD_MBR_1_ch
DISP_SD_MBR_2_FR_ch
DISP_SD_MBR_2_BA_ch
DISP_SD_MBR_3_ch
DISP_BARR_1_ch
DISP_BARR_2_ch
DISP_BUMPER_R_ch
DISP_BUMPER_L_ch
BARR_BT_ch
BARR_TP_ch
BARR_MID_ch
PULSE_SM36_ch
PULSE_SAE6O_ch
BARR_BT_SAE6O_ch
BARR_TP_SAE6O_ch

Do we have an innate sense for designing Information Visualizations?

Encoding Example Car Dataset

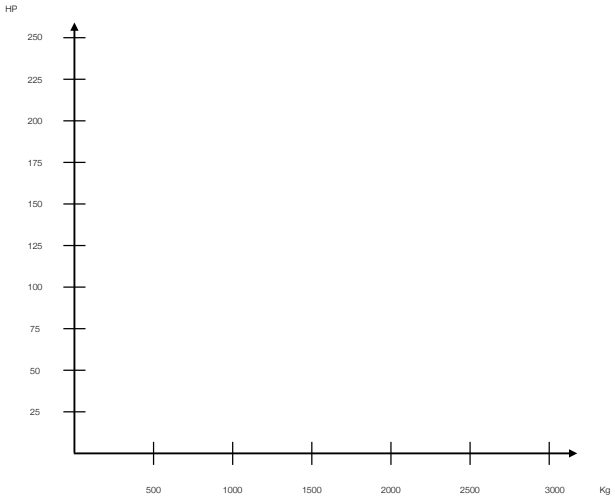
Brand	Name	Seats	Weight	Color	HP
	Smart	2	730	green	45
Renault	Twingo	4	790	petrol	59
Renault	Captur	5	1190	Yellow	66
VW	Golf	5	1304	Black	110
VW	Passat	5	1780	Silver	174
Ford	S Max	7	1980	Blue	180
Mercedes	E Class	5	1720	Red	255
Mercedes	V Class	7	2145	Gray	176
VW	T6 Multivan	8	2628	Black	150

Encoding Example Car Dataset



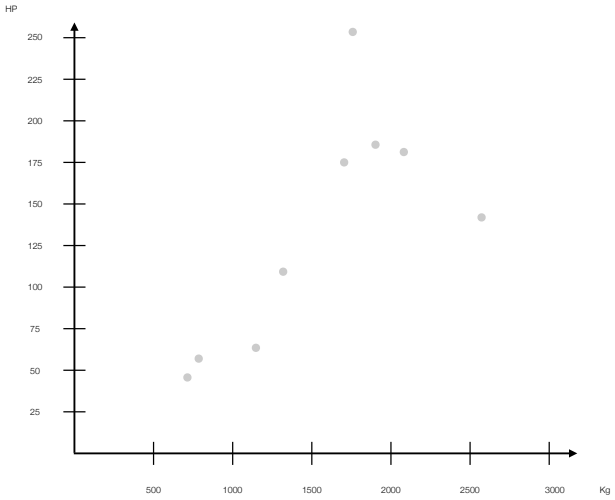
Brand	Name	Seats	Weight	Color	HP
	Smart	2	730	green	45
Renault	Twingo	4	790	petrol	59
Renault	Captur	5	1190	Yellow	66
VW	Golf	5	1304	Black	110
VW	Passat	5	1780	Silver	174
Ford	S Max	7	1980	Blue	180
Mercedes	E Class	5	1720	Red	255
Mercedes	V Class	7	2145	Gray	176
VW	T6 Multivan	8	2628	Black	150

Encoding Example Car Dataset



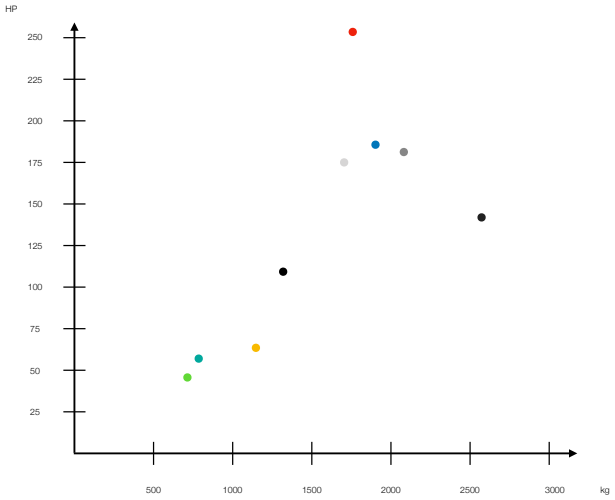
Brand	Name	Seats	Weight	Color	HP
	Smart	2	730	green	45
Renault	Twingo	4	790	petrol	59
Renault	Captur	5	1190	Yellow	66
VW	Golf	5	1304	Black	110
VW	Passat	5	1780	Silver	174
Ford	S Max	7	1980	Blue	180
Mercedes	E Class	5	1720	Red	255
Mercedes	V Class	7	2145	Gray	176
VW	T6 Multivan	8	2628	Black	150

Encoding Example Car Dataset



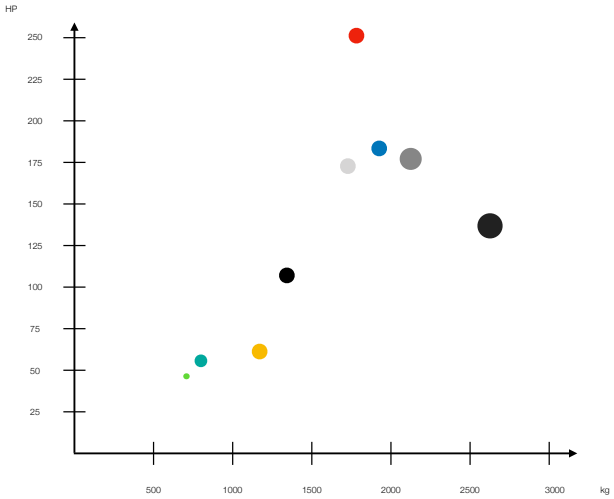
Brand	Name	Seats	Weight	Color	HP
	Smart	2	730	green	45
Renault	Twingo	4	790	petrol	59
Renault	Captur	5	1190	Yellow	66
VW	Golf	5	1304	Black	110
VW	Passat	5	1780	Silver	174
Ford	S Max	7	1980	Blue	180
Mercedes	E Class	5	1720	Red	255
Mercedes	V Class	7	2145	Gray	176
VW	T6 Multivan	8	2628	Black	150

Encoding Example Car Dataset



Brand	Name	Seats	Weight	Color	HP
	Smart	2	730	green	45
Renault	Twingo	4	790	petrol	59
Renault	Captur	5	1190	Yellow	66
VW	Golf	5	1304	Black	110
VW	Passat	5	1780	Silver	174
Ford	S Max	7	1980	Blue	180
Mercedes	E Class	5	1720	Red	255
Mercedes	V Class	7	2145	Gray	176
VW	T6 Multivan	8	2628	Black	150

Encoding Example Car Dataset



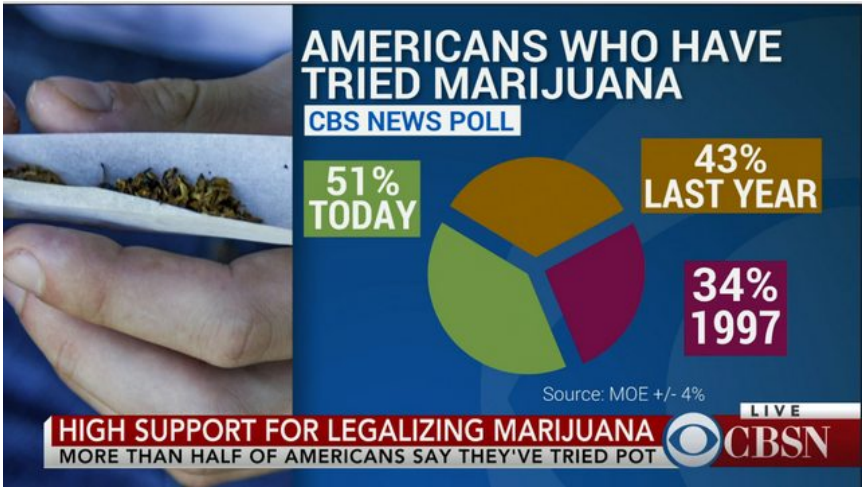
Brand	Name	Seats	Weight	Color	HP
	Smart	2	730	green	45
Renault	Twingo	4	790	petrol	59
Renault	Captur	5	1190	Yellow	66
VW	Golf	5	1304	Black	110
VW	Passat	5	1780	Silver	174
Ford	S Max	7	1980	Blue	180
Mercedes	E Class	5	1720	Red	255
Mercedes	V Class	7	2145	Gray	176
VW	T6 Multivan	8	2628	Black	150

Do we have an innate sense for designing Information Visualizations?

We will see next week!

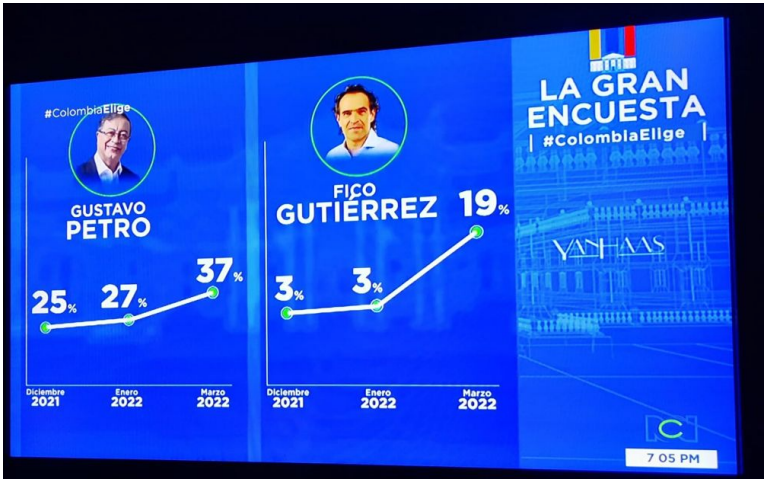
Bad Practices

Scale does not add up



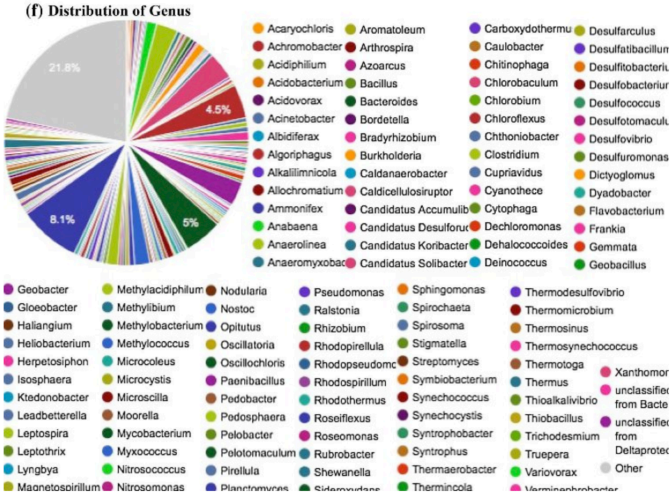
https://i0.wp.com/www.codeconquest.com/wp-content/uploads/image-9.png?w=640&ssl=1

Wrong alignment



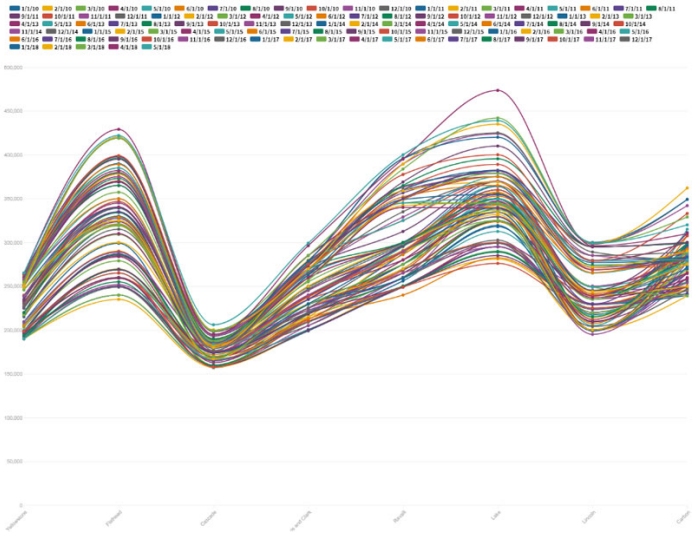
https://i0.wp.com/www.codeconquest.com/wp-content/uploads/image-10.png?w=1000&ssl=1

Wrong Chart Type and too many Colors

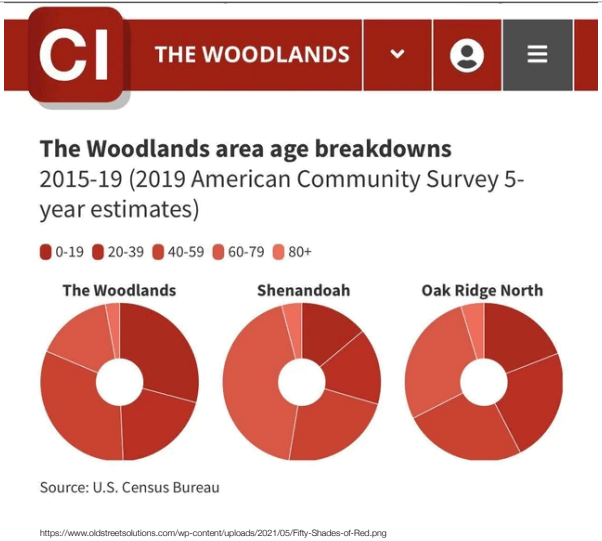


https://www.google.com/url?sa=i&url=https://humansofdata.atlan.com/2016/09/common-mistakes-bad-data-visualization/&psig=AOvVaw0cufzfIc8KQ1Y1WpWDb0wM&ust=1699265165675000&source=images&cd=vfe&ved=0CBEQjRxqFwoTCOjy-LmarYIDFQAAAAAdAAAAABAE

Wrong Chart Type and too many Colors



Poor Color Usage (Fifty Shades of Red)

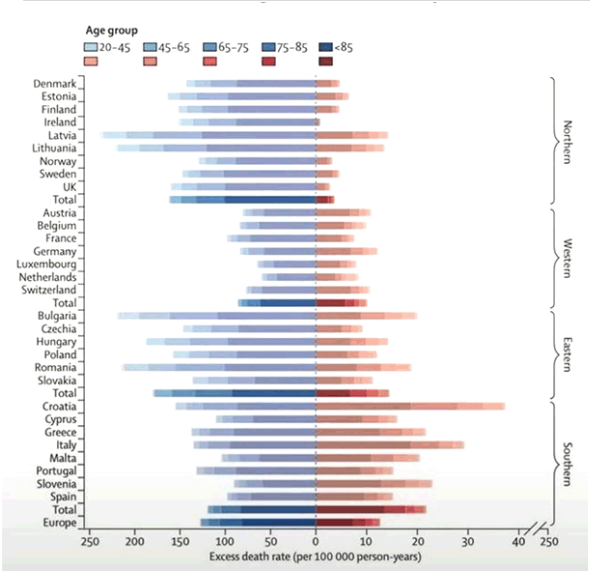


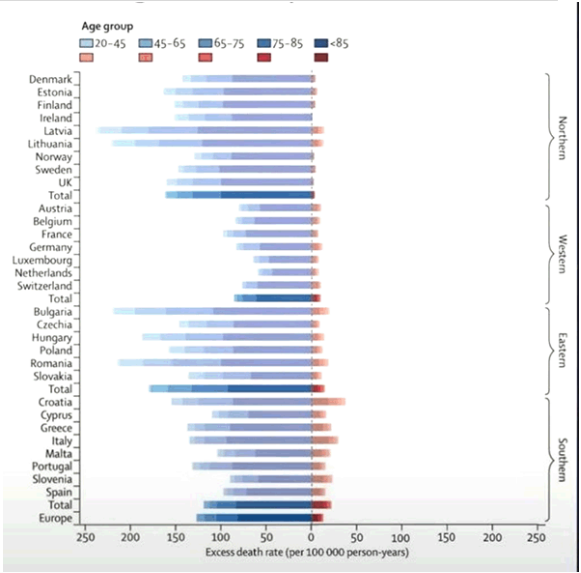
https://www.oldstreetsolutions.com/wp-content/uploads/2021/05/Fifty-Shades-of-Red.png

Truncated Axis

https://assets-global.website-files.com/61488f4f65be16b6ebbd450b/6216bbbe746ea37175559954_B0bgJV5rdHcVec3O-sLC2_EWOexj16ZLHGXe6WfNjf3dJfG4b6C5eYRRjA-Bns_Z8KkQHkRDLbhBy9ghGLBTTC3QWONg5fVBCRl9IcBCbl6V_dbZObsAXiUumcpim3c2FT2yh1yVj.png

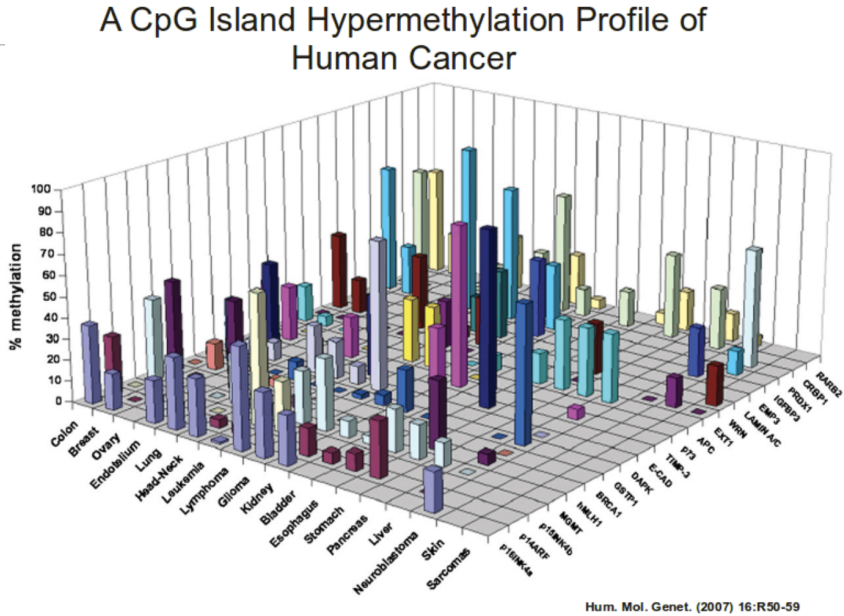
Tampered Axis





No 3D at all

- Don't do it!



https://www.oldstreetsolutions.com/wp-content/uploads/2021/05/3D-Chart.png

Best Practices

- Choose the most Appropriate Medium for Data Visualization depending upon the Objective it is meant to serve
- Provide necessary Context around Visuals
- Ensure Simplicity and Clarity of Information
- Where possible, bring in Originality by relating, seemingly Unrelated data and subjects
- Ensure Brevity and Avoid Unnecessary Information
- Use Simple and Easy to Understand Color Palettes
- Pay attention to Graphics in order to make sure that they are Visually Appealing

https://www.syntaxtechs.com/blog/data-visualization-examples

To Avoid

- Avoid using Too Many Variables within a single image which might result in distracting the viewers
- Be extremely careful of not visualizing data through an Unsuitable or Incorrect visualization format
- While using Scales in Data Visualization in order to depict differences between data points, it is important to ensure that the Scale is not Inconsistent
- Poor Choice of Colors is another significant issue which should be avoided at all costs. Thus, it is important to:
- Avoid using too many colors and colors with negligible contrast
- Avoid using conventional colors to convey opposite meanings
- Pay heed to the needs of people who might be colorblind

https://www.syntaxtechs.com/blog/data-visualization-examples

Thank You